

Compositional Operator Networks with Structured Latent Spaces

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Abstract

The document lays out a mathematical framework for operator-learning surrogates of parametrised, geometry-dependent, space–time fields, typically turbulent flows in which the learned latent space is explicitly compositional, disentangled, and physically interpretable. The treatment is self-contained and pedagogical: each machine-learning idea is introduced through its analogue in fluid mechanics, proper orthogonal decomposition (POD), dynamic mode decomposition (DMD), Galerkin methods, adjoint-based optimisation, or multidisciplinary design optimisation (MDO). Specific attention is given to (i) the practical realities of training on mixed DNS, LES, and experimental data that are gappy, noisy, and multi-resolution; (ii) the role of pre-learning transformations (Fourier, wavelet, POD, wall-scaling) and dual-head architectures that exploit them; (iii) a survey of operator-network architectures (DeepONet, FNO, CNO, GNO, transformers, neural fields, diffusion) with pros and cons; and (iv) an extensive library of physics-motivated loss functions and their multi-objective interpretation. The framework builds on recent work by our group (Powadi et al., 2025; Guo et al., 2016; Rabeh et al., 2025b; Karki et al., 2025a; Nirmal et al., 2025; Tali et al., 2025; Rabeh et al., 2025a; Karki et al., 2025b).

How to read this document

The intended reader is an expert in fluid mechanics, MDO, or computational PDE methods, who is comfortable with POD, Galerkin projection, adjoint methods, and surrogate modelling, but who has not built neural networks.

Each major idea appears in three layers:

- **Main text** — the mathematical statement, in the vocabulary of operator learning.
- **Fluid-mechanics reading** (*orange box*) — the same idea in the language of turbulence, wall-bounded flows, or PDE theory.
- **MDO / computational-science reading** (*green box*) — connections to surrogate modelling, coupled-disciplinary problems, data assimilation, or co-Kriging.
- **What this is, in ML terms** (*blue box*) — when the concept is foreign to classical PDE work, a concise ML-literacy gloss.

The document has six parts. Part I sets foundations. Part II addresses data, including mixed DNS + experimental inputs, gappy measurements, and pre-learning transformations. Part III surveys operator-network architectures. Parts IV and V develop the framework proper — the compositional latent space, the loss library, the multi-objective and multidisciplinary viewpoints. Part VI treats use and open questions.

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Part I

Foundations

1 A Rosetta stone: machine-learning terms in fluid-mechanics vocabulary

ML term	Closest fluid-mechanics / computational analogue	One-line meaning
Neural network	Nonlinear basis expansion with fitted coefficients	Stack of affine maps and element-wise nonlinearities
Weights θ	Coefficients / parameters in the surrogate	Numbers adjusted during fitting
Training	Fitting, calibration, regression	Minimising a loss over parameters
Loss function	Objective functional, residual	Scalar whose minimiser defines the fit
Gradient descent	Iterative optimiser (quasi-Newton, etc.)	Repeated parameter update along $-\nabla \mathcal{L}$
Backpropagation	Reverse-mode adjoint (as in adjoint CFD)	Efficient gradient of $\mathcal{L}$ w.r.t. all θ
Autoencoder	POD + truncation + reconstruction (nonlinear)	High-dim field $\rightarrow$ low-dim code $\rightarrow$ field
Encoder $\mathcal{E}$	Projection onto modes	$\mathbf{U} \mapsto \mathbf{z}$ (low-dim coordinates)
Decoder $\mathcal{D}$	Synthesis from modes	$\mathbf{z} \mapsto \mathbf{U}$ (field reconstruction)
Latent space $\mathbf{z}$	Modal amplitudes, reduced coordinates	Low-dim code for one snapshot / trajectory
Operator network	Learned solution map or Green's function	$(\boldsymbol{\mu}, \mathbf{g}) \mapsto \mathbf{U}(\cdot, \cdot)$
Embedding	Feature, learned coordinate	Vector summarising an input
Branch / trunk (DeepONet)	Input-amplitudes / basis-functions of a Galerkin scheme	Two networks whose dot product gives the field
Fourier neural operator (FNO)	Spectral (Fourier–Galerkin) PDE discretisation	Learn the convolution kernel in Fourier space
Attention / transformer	Non-local operator with data-dependent kernel	Content-dependent weighting over tokens
Graph neural network (GNN)	Finite-volume stencil / mesh-node update	Local message passing on a graph
Physics-informed loss	PDE-residual penalty, variational formulation	Include PDE residual in the loss
Multi-task learning	Multi-objective calibration	Several losses combined during training
Transfer learning	Warm-starting / continuation / homotopy	Reuse weights from a related problem
Disentanglement	Orthogonalisation of factors (PCE / Sobol)	Independent axes in the latent
Exposure bias	Error accumulation under autoregressive rollout	Trained on true inputs but deployed on its own outputs
Diffusion model	Iterative denoising / refinement generator	Sample noise, progressively denoise toward data

A few terms deserve a little more detail.

“Operator” in operator learning. In the PDE literature an operator is a map between function spaces, e.g. the heat semigroup $S_t : u_0 \mapsto u(\cdot, t)$, or the solution map $(f, g) \mapsto u$ of a BVP. Neural operators aim to learn these maps. The novelty with respect to classical surrogates (kriging, radial basis functions) is that the input *and* output are genuinely function-valued, not finite-dim vectors. Universal-approximation theorems for operators (Chen and Chen, 1995; Lu et al., 2021; Kovachki et al., 2023) underwrite the approach.

“Latent space” and nonlinear POD. A POD-truncated reduced-order model writes

$$U(\mathbf{x}, t) \approx \sum_{i=1}^r a_i(t) \phi_i(\mathbf{x}), \quad \phi_i \text{ POD modes, } a_i(t) \text{ amplitudes.}$$

An autoencoder generalises this to a *nonlinear* reduced basis: $U \mapsto \mathbf{z} \in \mathbb{R}^r$ by a nonlinear encoder, $\mathbf{z} \mapsto \tilde{U}$ by a nonlinear decoder. The same r -dimensional bottleneck now captures phenomena that linear POD cannot — moving shocks, advecting fronts, symmetry-breaking bifurcations — all of which frustrate linear-mode truncation. The rest of this document can be read as: *build a nonlinear reduced-basis model whose reduced coordinates $\mathbf{z}$ have identifiable physical meaning.*

“Training” is regression. The pervasive ML term “training a model” means, literally, *solve the regression problem* $\min_{\theta} \mathcal{L}(\theta; \text{data})$. The only differences from classical nonlinear regression are scale (millions of parameters) and the use of reverse-mode automatic differentiation to get $\nabla_{\theta} \mathcal{L}$ efficiently. The computational kernel is identical to an adjoint-based optimisation in PDE-constrained design.

MDO / computational-science reading

Note that the gradient $\nabla_{\theta} \mathcal{L}$ needed for training is exactly what an adjoint solve produces: the sensitivity of an output functional with respect to all parameters, at cost independent of the number of parameters. “Backpropagation” is reverse-mode automatic differentiation, which is how adjoint codes are implemented when the PDE discretisation is written in an AD-compatible framework (e.g. FEniCS, JAX, PyTorch). The forward pass of a network is the primal; backpropagation is the adjoint.

2 Problem setting

Let $\Omega_{\mathbf{g}} \subset \mathbb{R}^d$, $d \in \{2, 3\}$, be a spatial domain whose shape is encoded by a geometric descriptor $\mathbf{g} \in \mathcal{G}$. Two equivalent descriptions of $\mathcal{G}$ are admitted:

- **Finite-dimensional parameterisation:** $\mathbf{g} = (g_1, \dots, g_q) \in \mathbb{R}^q$ — airfoil (CST or Bernstein) coefficients, NURBS control points, or the vector of an airfoil-family design variables.
- **Implicit representation:** $\mathbf{g}$ encoded as a signed distance function $\phi_{\mathbf{g}} : \mathbb{R}^d \rightarrow \mathbb{R}$, with $\Omega_{\mathbf{g}} = \{\mathbf{x} : \phi_{\mathbf{g}}(\mathbf{x}) < 0\}$. This is the Karki et al. (2025a); Rabeh et al. (2025b) convention; it supports continuous queries, handles non-parametric shapes, and differentiates cleanly.

Let $\boldsymbol{\mu} \in \mathcal{P} \subset \mathbb{R}^p$ denote operating parameters: (Re, M , roughness amplitude, inflow turbulence intensity, wall temperature, forcing amplitude, ...). The space–time field of interest is

$$U : \Omega_{\mathbf{g}} \times [0, T] \times \mathcal{P} \times \mathcal{G} \longrightarrow \mathbb{R}^c,$$

with c the number of physical channels — e.g. (u, v, w, p) for incompressible flow ($c = 4$), $(\rho, \rho \mathbf{u}, \rho E)$ for compressible ($c = 5$), richer if including temperature, species, or turbulence-model variables.

The *solution operator* is

$$\mathcal{S} : \mathcal{P} \times \mathcal{G} \longrightarrow L^2(\Omega_{\mathbf{g}} \times [0, T]; \mathbb{R}^c), \quad (\boldsymbol{\mu}, \mathbf{g}) \mapsto \mathbf{U}(\cdot, \cdot; \boldsymbol{\mu}, \mathbf{g}). \quad (1)$$

Objective. Build a differentiable, fast-to-query surrogate $\hat{\mathcal{S}}_{\Theta} \approx \mathcal{S}$ that passes through a *structured* latent space — one whose coordinates carry identifiable physical meaning — and can be used downstream for design optimisation, UQ, data assimilation, or symbolic discovery.

Fluid-mechanics reading

Think of $\mathcal{S}$ as the “all-at-once” answer to: *given a shape and an operating condition, what is the full flow field over all space and time?* A direct CFD evaluation of $\mathcal{S}$ at one $(\boldsymbol{\mu}, \mathbf{g})$ is expensive; we want a cheap surrogate. POD+Galerkin, DMD, or kriging on QoI’s are classical answers. Operator learning is the modern equivalent at the level of entire space–time fields.

3 The core idea, at one page

A single paragraph would say: train an encoder–propagator–decoder triple $(\mathcal{E}, \Phi, \mathcal{D})$ in which the encoder compresses field snapshots into a compact latent vector, the propagator advances that latent vector in time, and the decoder reconstructs fields from it:

$$U(\cdot, 0) \xrightarrow{\mathcal{E}} \mathbf{z}_{\text{dyn}}(0) \xrightarrow{\Phi_{\Delta t}} \mathbf{z}_{\text{dyn}}(\Delta t) \xrightarrow{\Phi_{\Delta t}} \dots \xrightarrow{\mathcal{D}} U(\cdot, t),$$

with operating parameters $\boldsymbol{\mu}$ and geometry $\mathbf{g}$ injected through a separate static encoder whose output does not evolve in time. The remainder of this section expands the three non-obvious ingredients — the block-structured latent, the mechanism by which disentanglement is induced, and why this matters for downstream scientific work.

The vocabulary of disentanglement

Disentanglement is the hypothesis, articulated clearly by Bengio et al. (2013) and pursued as a central research program ever since, that high-dimensional natural data are generated by a small number of independent, interpretable factors of variation, and that these factors can in principle be recovered by learning. For images of faces, the factors might be identity, pose, lighting, expression, skin tone; for our setting — space–time fields of turbulent flows — the factors are the regime variables (Re, M) , the geometric shape, the boundary and forcing conditions, and the stochastic turbulent state. A *disentangled representation* is one in which each factor is encoded by its own coordinate (or its own block of coordinates) in a latent vector, such that moving along that coordinate changes exactly one semantic factor while leaving the others fixed. This is the representational dream of factorisable models.

The ML literature has pursued this dream aggressively in vision. β -VAE (Higgins et al., 2017) penalises the KL divergence between the encoder’s posterior and a factorised prior, pressuring the latent bottleneck to factorise. β -TCVAE (Chen et al., 2018a) decomposes the evidence lower bound (ELBO) to isolate a total-correlation term that directly measures the dependence among latent coordinates, and penalises it. InfoGAN (Chen et al., 2016) maximises mutual information between a designated subset of latent codes and the generated output, so those codes must carry interpretable information. StyleGAN (Karras et al., 2019) organises its latent space hierarchically so that coarse-to-fine scales control coarse-to-fine visual attributes — pose, then identity, then texture. In multimodal learning, CLIP (Radford et al., 2021) aligns image and language embeddings in a shared space by contrastive pretraining, with the linguistic side implicitly factorising the visual representation along semantic axes.

The single most important result of this line of work is negative. Locatello et al. (2019) proved, and confirmed over twelve thousand trained models, that *disentanglement is impossible from unlabelled data without inductive bias on both the model and the data-generating process*. Without such bias, any encoder–decoder pair that reconstructs well admits continuous families of equally good reconstructions that shuffle the factors (non-linearly) among the latent coordinates. There is no reason for the axes of $\mathcal{Z}$ to align with the true generative axes unless one forces them to.

How we induce disentanglement here

Our inductive bias is structural and, for scientific data, natural. It rests on three levers.

First, a block-structured latent: we fix, by architecture, that $\mathbf{z} = [\mathbf{z}_{\mu} \parallel \mathbf{z}_{\mathbf{g}} \parallel \mathbf{z}_{\eta} \parallel \mathbf{z}_{\xi}]$ decomposes into four blocks corresponding to operating parameters, geometry, coherent dynamics, and residual

fluctuations (This is a direct generalisation of the $\mu/g/\xi$ split of our prior work [Powadi et al. \(2025\)](#)). Architectural separation is necessary but not sufficient: without further constraints, the four blocks can scramble information arbitrarily.

Second, group-structured minibatching: we curate minibatches in which samples share factors. Every minibatch contains groups of samples at the same μ with different g , and groups at the same g with different μ . The fusion block takes all encoded samples within a group and extracts a shared sub-code. This is formally analogous to the auxiliary-variable construction that makes iVAE ([Khemakhem et al., 2020](#)) identifiable — the group label plays the role of the auxiliary variable u — and converts an impossibility result into a provable one in the idealised case.

Third, loss-level pressure for block independence: we penalise Pearson correlation ([Powadi et al., 2025](#)) and, for nonlinear dependence, HSIC ([Gretton et al., 2005](#)) between blocks; we use contrastive and swap-consistency losses to make the compositional structure functional rather than merely statistical. Crucially, our scientific data-generating process *actually has* the factorised structure these methods presume, because μ and g are physically independent control knobs — unlike the case for natural images, where the putative generative factors are an epistemological convenience.

Why disentanglement matters for fluid mechanics and design

Three concrete payoffs distinguish disentangled latent spaces from black-box surrogates:

Interpretable design-space exploration. Sweeping z_μ with z_g fixed traces a curve of operating conditions through fluid-field space, at fixed geometry; sweeping z_g with z_μ fixed traces a shape-family curve at fixed regime. Each sweep is a well-defined scientific experiment. In an entangled latent, the equivalent sweep changes both and is uninterpretable. MDO benefits from the same property: the design parameterisation is intrinsically orthogonal, and classical ill-conditioning of correlated design variables is avoided by construction.

Cheap explainable dynamical models and closures. A propagator Φ that acts on z_{dyn} and leaves z_{stat} untouched is a dynamical system parameterised by (μ, g) . Applying sparse identification ([Brunton et al., 2016](#); [Champion et al., 2019](#)) to the latent trajectories yields equations of the form

$$\frac{dz_\eta}{dt} = \sum_j c_j(\mu, g) \Theta_j(z_\eta),$$

in which the coefficients depend *cleanly* on regime because the regime variables live in a separate, decorrelated block. In an entangled latent, c_j would implicitly depend on initial-condition seed and geometry residual, the regression would be ill-posed, and the discovered equations would overfit. The residual block z_ξ serves a second purpose: a learned map $z_\xi(t + \Delta t) \mid z_\eta(t), \mu$ could be interpreted as a *subgrid closure*, data-driven and trained directly against observed fluctuation statistics — the latent-space analogue of LES and RANS closure modelling.

Transfer, extrapolation, and symbolic checks. Because z_μ is an identified axis, extrapolation along it to new Reynolds or Mach numbers is well-defined rather than a dead-reckoning leap; the propagator’s (μ, g) -conditioning is explicit rather than hidden. In tractable limits (low-Re, incompressible) the symbolic equations extracted from the latent dynamics must reduce to textbook results — a built-in falsifiability check that an entangled surrogate cannot offer.

What this is, in ML terms

Everything above is “just” a carefully structured autoencoder-plus-propagator trained on a composite loss with stochastic gradient descent (Adam or similar). The interesting part is the structural inductive bias — block architecture, group-batched data, independence penalties, factor-labelled supervision — not the training dynamics, which are stock. The gap between vision-style disentanglement (where the factors are a modelling assumption about natural images) and physics-style disentanglement (where the factors are *literally* the control knobs of the experiment) is why we expect the scientific setting to be the one where this finally works well.

4 Why compositionality in physics is easier than in computer vision

We claim that fluid mechanics is a more hospitable setting for disentanglement than computer vision. This section makes that claim precise. The argument has four strands, each of which removes one of the specific obstacles that the vision-side disentanglement literature has struggled against.

Dimensional analysis already gives us the axes

Vision-style disentanglement must *discover* its factors — identity, pose, lighting — from unlabelled data, and the Locatello impossibility result (Locatello et al., 2019) tells us this is hopeless without supervision. Physics has a theorem that provides the factors by fiat. Buckingham’s Π -theorem (Buckingham, 1914) states that if n physical variables describe a phenomenon and involve k independent physical dimensions, the phenomenon can be rewritten as a relationship among exactly $n - k$ dimensionless groups. For aerodynamic flow around a body of characteristic length L at free-stream speed U in a fluid of density ρ and viscosity μ , the governing dimensionless groups are Reynolds Re , Mach M , and geometric ratios g ; for heat transfer we add Prandtl Pr ; for rotating systems we add Rossby Ro ; and so on (Pope, 2000; Barenblatt, 1996). These are not conjectured from data — they are derived from dimensional consistency of the Navier–Stokes equations, and they have been organising fluid mechanics for a century.

The practical consequence is direct. In vision, the block structure $\mathbf{z} = [\mathbf{z}_{\text{identity}} \parallel \mathbf{z}_{\text{pose}} \parallel \mathbf{z}_{\text{lighting}} \parallel \dots]$ is a modelling choice that the data may or may not admit. In our setting the block structure $\mathbf{z} = [\mathbf{z}_{\mu} \parallel \mathbf{z}_g \parallel \mathbf{z}_{\eta} \parallel \mathbf{z}_{\xi}]$ with $\boldsymbol{\mu} = (Re, M, \dots)$ has a closed-form derivation: the regime axis is exactly the set of Pi-groups that dimensional analysis identifies. We are not hoping the data factorise along our chosen axes; the governing equations already guarantee it, modulo the parametrisation of $\mathbf{g}$.

Dynamic similarity guarantees that factor sweeps are meaningful

Disentanglement claims its full force when one can independently sweep each factor and verify that the flow responds as expected. In vision this is ill-defined: fixing identity while varying pose is an abstraction that has no independent ground truth, because real humans cannot be physically decoupled into identity and pose. In fluid mechanics this is trivially well-defined, and has been the engineering practice for over a century. The principle of *dynamic similarity* states that two flows with the same Pi-groups are scaled copies of each other — a wind-tunnel model at matched Reynolds number is, in the relevant sense, the same flow as the full-scale prototype. Sweeping $\mathbf{z}_{\mu}$ at fixed $\mathbf{z}_g$ corresponds to tracing a line through the space of dimensionless parameters, and there

are physically realisable flows at every point on that line. For any predicted behaviour along the sweep, one can in principle perform a matched wind-tunnel experiment and check.

Fluid-mechanics reading

This is why engineers have been able to scale lift, drag, and heat-transfer coefficients between models and prototypes since the Wright brothers. Disentanglement, under the name of dimensional-similarity scaling, has been the quiet workhorse of aerospace and naval architecture. What we are adding is a representation-learning layer that absorbs the same structure into a neural surrogate.

Universality classes give us a finite menu of behaviours

A deeper structural gift from turbulence theory is universality. The Kolmogorov K41 description of the inertial range (Frisch, 1995; Pope, 2000) predicts that sufficiently high-Reynolds-number turbulent flows, at sufficiently small scales and away from boundaries, display universal statistical structure: the energy spectrum scales as $E(k) \sim \varepsilon^{2/3} k^{-5/3}$, the second-order structure function scales as $(\varepsilon r)^{2/3}$, and these properties do not depend on the details of the large-scale forcing. Universality is well-understood to be approximate (intermittency corrections to K41 exist, and are active research in their own right), but the zeroth-order picture — a small *menu* of universal statistical behaviours, indexed by regime — is robust and experimentally supported.

For representation learning this is a gift. In vision, the space of plausible latent factorisations is combinatorial: a face can be factored into identity-plus-pose-plus-lighting, or into shape-plus-texture-plus-illumination, or into symmetric-plus-asymmetric components, with little in the data to select between them. In turbulence, the residual fluctuation field $\mathbf{z}_\xi$ does not have the degrees of freedom that vision-style disentanglement would grant it: at inertial-range scales it must obey K41 statistics (or documented departures from them), and at the regime-dependent level it inherits a small number of universal distributions indexed by μ . The latent-space capacity of $\mathbf{z}_\xi$ can therefore be *bounded* by physical theory rather than guessed, and the spectral-matching losses of §15 (L 22, L 23) pressure the network toward the universal statistics rather than drifting toward whatever the reconstruction objective makes convenient.

Factors are physical control knobs, not abstractions

The deepest distinction is epistemological. In vision, “identity” and “pose” are human-interpretable summaries of phenomena that the generative process does not actually instantiate as separate variables — a photograph of a person is a projection from a high-dimensional physical process (lighting, geometry, camera optics, skin scattering) that does not itself factorise into identity and pose. Disentanglement here is a human-perceptual convenience, not a property of the data-generating process, and the impossibility result of Locatello et al. (2019) applies with full force.

In the experiments we actually run — Iowa State’s tabletop turbulence simulator, our high-Reynolds wall-bounded DNS, our PIV of separated flows — the factors *are* the physical control knobs. Setting Re is adjusting a pump. Setting M is adjusting a stagnation pressure. Setting $\mathbf{g}$ is swapping a model geometry. These are independently controllable, and the dataset is typically acquired by varying them on a designed grid; group-labelled minibatching of the kind we use is simply asking the loss function to respect the already-existing experimental structure. The data-generating process factorises because the experimenter factorised it.

MDO / computational-science reading

This is the point of strongest contact with MDO. A multidisciplinary design landscape is *defined* by its controllable parameters; design space is by construction a Cartesian product of factor axes. A disentangled surrogate aligns its internal representation with this product structure automatically, removing the ill-conditioning that correlated design variables typically produce. In effect, dimensional analysis + experimental design + disentangled-representation learning compose into a closed loop in which each component is already used in industrial practice — we are formalising an existing workflow, not inventing a new one.

The caveats worth keeping in mind

None of the above is a guarantee. Three qualifications are needed:

Regime transitions violate smooth compositionality. Laminar-to-turbulent transition, subsonic-to-supersonic transition, attached-to-separated flow, and bifurcations more generally are points where the flow changes qualitatively and the smooth manifold structure assumed by a neural decoder breaks down. Across such transitions, swap-consistency (L 12) may fail and the block-factorisation will need to accommodate discrete regime labels in addition to continuous axes. This is a research question.

Near-transonic and strongly-nonlinear regimes break HSIC orthogonality. At $M \approx 1$, drag coefficient depends on both Re and M in a strongly coupled way; shock-boundary-layer interaction is intrinsically non-factorisable at the local level. Hard orthogonality (L 6, L 8) is too strong in these regimes; HSIC (L 7) and soft block-Gram penalties (L 9) are the appropriate alternatives. The framework accommodates this by letting the disentanglement losses be weighted down adaptively where the physics genuinely couples the factors.

Scale-separation assumptions are idealisations. K41 universality is strongest at high Reynolds numbers and for statistics far from walls. Wall-bounded, roughness-coupled, and moderate-Reynolds flows — which are exactly the regimes of interest for hypersonic boundary layers (Raje et al., 2025) and many industrial problems — show measurable departures. The framework accommodates this by letting z_ξ carry the regime-dependent departure from universality, rather than assuming universality is exact.

The net claim, carefully stated: the scientific setting gives us *axes to align to* (Pi-groups), *tests we can run* (dynamic-similarity experiments), *statistical targets for the residual* (K41 and its documented departures), and *data generated by a genuinely factorised process* (experimental design). Each removes one obstacle the vision literature has had to work around. The remaining open problems are real but localised, and they do not require us to abandon the structural picture — only to be careful where it bends.

Data, measurements, and pre-learning transformations

5 The data landscape: DNS, LES, RANS, experiments

Our training data are heterogeneous. A realistic corpus for turbulent-flow operator learning contains at least four kinds:

1. **Direct numerical simulation (DNS).** All scales resolved down to the Kolmogorov length $\eta_K \sim (\nu^3/\varepsilon)^{1/4}$. Regular grid or spectral. High fidelity, expensive, feasible only at moderate Reynolds numbers and simple geometries. Output: (u, v, w, p) on a dense grid, at many timesteps. Usually clean but potentially under-sampled in time relative to the Kolmogorov time τ_K .
2. **Large-eddy simulation (LES).** Large scales resolved, small scales modelled by a subgrid closure. Regular grid. Medium fidelity, medium cost, broader Re range. Output similar to DNS but filtered; the residual subgrid stress is a *hidden variable* whose effect on resolved fields is baked in through the closure.
3. **Reynolds-averaged Navier–Stokes (RANS/URANS).** Time-mean (or ensemble-mean) fields with a turbulence closure. Unstructured mesh, industrial-scale geometries, cheapest. Only first- and second-order statistics; no instantaneous structures. The closure model is a source of systematic bias.
4. **Experiments.** Particle-image velocimetry (PIV), laser-Doppler velocimetry (LDV), hot-wire anemometry, pressure-sensitive paint (PSP), surface-mounted pressure taps, schlieren or Mach–Zehnder for compressible; LIF for scalars. Output is almost always *gappy*: 2D planes from PIV, point time series from hot wires and pressure taps, integral quantities from loads, and so on. Noise levels are non-trivial and often spatially heterogeneous (PIV errors grow at low signal-to-noise regions).

Any realistic AutoDIDACTS dataset — especially one that crosses into tabletop turbulence simulator data — will mix all four. This creates three practical problems: (i) data are on different grids, (ii) data have different coverage, (iii) data have different fidelities. Each must be handled explicitly in the loss and in the encoder design.

MDO / computational-science reading

This is the same situation one faces in multi-fidelity surrogate modelling: a few high-fidelity points, many low-fidelity points, perhaps some intermediate. Classical answers — co-Kriging, multi-fidelity Monte Carlo, hierarchical Gaussian processes — all rely on a fidelity hierarchy and on the assumption that low-fidelity models correlate usefully with high-fidelity ones. Operator-learning analogues exist and will be used (see §17.5).

6 Handling gappy, noisy, multi-resolution measurements

Three strategies, applied in combination.

6.1 Coordinate-based encoders: let the network live on the point cloud

The principal challenge with mixed-grid data is that one model cannot naturally consume both a 1024×256 structured grid from DNS and a few thousand scattered PIV vectors from an experiment. The solution is to represent every field as a *function of coordinates*, rather than as an array on a fixed grid:

$$\text{snapshot} = \{(\mathbf{x}_m, \mathbf{U}_m)\}_{m=1}^{M_i}, \quad M_i \text{ varies across samples.}$$

The encoder is then a function of the point cloud. Two practical architectures:

PointNet-style encoders (Qi et al., 2017). A shared MLP is applied to each $(\mathbf{x}_m, \mathbf{U}_m)$ individually, then a permutation-invariant pooling (max or mean) aggregates into a single vector:

$$\mathcal{E}(\{(\mathbf{x}_m, \mathbf{U}_m)\}) = \text{Pool}(\{\text{MLP}(\mathbf{x}_m, \mathbf{U}_m)\}_m).$$

Permutation invariance mirrors the fact that the *set* of measurements, not their ordering, defines the snapshot.

Implicit neural representation (INR) decoders. The decoder becomes a function $\mathcal{D}(\mathbf{z}; \mathbf{x}, t)$ of the query location — not an array. This is what DeepONet’s trunk does (Lu et al., 2021), and it is what Karki et al. (2025a) leverage for SDF-based geometry. Querying at any density is then free.

Together, coordinate-based encoders and INR decoders decouple the model from any particular grid. Training batches can include samples at wildly different resolutions, with no padding, interpolation, or projection onto a common mesh.

6.2 Masked losses: don’t penalise unobserved points

When data are gappy, the reconstruction loss must be evaluated only at observed locations:

$$\mathcal{L}_{\text{rec}}^{\text{masked}} = \mathbb{E} \left[\frac{1}{|\mathcal{O}_i|} \sum_{m \in \mathcal{O}_i} \|\mathbf{U}_m - \hat{\mathbf{U}}(\mathbf{x}_m, t_m; \mathbf{z})\|^2 \right], \quad (2)$$

where $\mathcal{O}_i$ is the observation set for sample i . This is the operator-learning analogue of *gappy POD* (Everson and Sirovich, 1995; Willcox, 2006): reconstruct the field from partial measurements by projecting onto a basis and finding the coefficients that best match observations at the sensor locations. Here the “basis” is nonlinear and learned.

6.3 Heteroscedastic weighting: respect the noise

Not all measurements are equally trustworthy. A PIV vector in a low-seeding region, a hot-wire near a stagnation point, or a DNS snapshot from a coarse simulation should all carry different weights. The principled way is to assume a data-generating model

$$\mathbf{U}_m = \mathbf{U}_m^{\text{true}} + \boldsymbol{\epsilon}_m, \quad \boldsymbol{\epsilon}_m \sim \mathcal{N}(0, \sigma_m^2 \mathbf{I}),$$

and to use the negative log-likelihood,

$$\mathcal{L}_{\text{het}} = \mathbb{E} \left[\sum_m \left(\frac{\|\mathbf{U}_m - \hat{\mathbf{U}}(\mathbf{x}_m, t_m)\|^2}{2\sigma_m^2} + \log \sigma_m \right) \right]. \quad (3)$$

The variances σ_m^2 can be (i) provided by the measurement instrument (uncertainty analysis), (ii) learned jointly with the network as an output head, or (iii) block-constant by data source.

Fluid-mechanics reading

Heteroscedastic weighting is the operator-learning analogue of Gauss–Markov weighted least squares, or of 3DVAR / 4DVAR data assimilation, in which the observation operator H projects model state to measurement space and observations contribute through $(H\mathbf{U} - \mathbf{U}^{\text{obs}})^\top R^{-1} (H\mathbf{U} - \mathbf{U}^{\text{obs}})$ with R the measurement covariance. Equation (3) is exactly this structure.

6.4 Cross-source consistency

When sample i exists in more than one data source (e.g. an experimental case that has also been simulated), impose

$$\mathcal{L}_{\text{src}} = \|\mathcal{E}^{\text{DNS}}(\mathbf{U}_i^{\text{DNS}}) - \mathcal{E}^{\text{exp}}(\mathbf{U}_i^{\text{exp}})\|^2, \quad (4)$$

or, less strictly, that their static embeddings agree. This loss ties the two encoders together in latent space and makes the surrogate interpret cross-source data consistently.

6.5 Bias correction

Experimental data may contain systematic bias (calibration drift in pressure taps, probe interference, wall-proximity effects in hot wires) that should not contaminate the learned operator. A small linear bias-correction head $\mathbf{U}_m^{\text{exp}} \leftarrow \mathbf{U}_m^{\text{exp}} - b(\boldsymbol{\mu}_i, \mathbf{g}_i)$ can be co-trained using a prior that $\mathbb{E}[b] = 0$, shifting the DNS “truth” and experimental observations toward mutual consistency without overfitting to a specific calibration.

7 Pre-learning transformations: choose the space in which to represent data

Representation dominates performance in turbulence modelling. A well-chosen transform of the input field before the network sees it can halve the parameter count and dramatically improve generalisation. This section surveys the relevant options.

7.1 Fourier transform

Definition. For a periodic domain, $\hat{\mathbf{U}}(\mathbf{k}) = \int_{\Omega} \mathbf{U}(\mathbf{x}) e^{-i\mathbf{k} \cdot \mathbf{x}} d\mathbf{x}$. In time, $\tilde{\mathbf{U}}(\omega)$.

When it helps.

- Periodic or near-periodic domains (homogeneous isotropic turbulence, channel flow in stream-wise/spanwise directions, homogeneous shear).
- Problems where the energy cascade in k -space is central: Kolmogorov $E(k) \sim k^{-5/3}$, Kraichnan in 2D, rotating/stratified modifications.

- Long-range spatial correlations: in Fourier space these become pointwise, which is cheap.

When it hurts. Wall-bounded flows with complex geometry: Fourier basis does not respect no-slip walls, so representing boundary layers in Fourier modes requires many high- k components and is wasteful. One partial remedy is “hybrid” decomposition: Fourier in homogeneous directions, Chebyshev or physical-space in wall-normal.

Fluid-mechanics reading

Fourier Neural Operators (FNO; Li et al., 2021) are built exactly on this transform: each FNO layer computes an FFT, multiplies by a learned kernel in k -space (low-pass truncated), then computes an inverse FFT. This is numerically identical to a Fourier–Galerkin discretisation of a convolution integral. The strength is computational: convolutions in physical space are pointwise multiplications in Fourier space, at $\mathcal{O}(N \log N)$ cost per layer.

7.2 Wavelet transform

Definition. A continuous wavelet transform $\hat{U}(a, \mathbf{b}) = \int \mathbf{U}(\mathbf{x}) \bar{\psi}_{a,\mathbf{b}}(\mathbf{x}) \, d\mathbf{x}$ with translated, scaled copies $\psi_{a,\mathbf{b}}(\mathbf{x}) = a^{-d/2} \psi((\mathbf{x} - \mathbf{b})/a)$. Discrete wavelet transforms use dyadic grids.

When it helps.

- Non-periodic, non-homogeneous flows.
- Intermittent events: wavelets localise in both position and scale.
- Near-wall turbulence and shock-dominated flows, where features are spatially concentrated.

When it hurts. Smooth global structures are represented by fewer Fourier modes; wavelets have redundancy.

Wavelet Neural Operators (WNO; Tripura and Chakraborty, 2023) replace FNO’s Fourier step with a wavelet step, trading global spectral efficiency for local adaptivity.

7.3 Proper orthogonal decomposition (POD)

Data-driven linear basis (Sirovich, 1987; Berkooz et al., 1993): $\phi_i(\mathbf{x})$ are eigenfunctions of the two-point correlation $R(\mathbf{x}, \mathbf{x}') = \mathbb{E}[\mathbf{U}(\mathbf{x})\mathbf{U}(\mathbf{x}')^\top]$. Use POD *offline* before the network:

$$\mathbf{U}(\mathbf{x}, t) \approx \sum_{i=1}^r a_i(t) \phi_i(\mathbf{x}), \quad \text{input to NN} = (a_1, \dots, a_r).$$

When it helps. Mild nonlinearity; training data that lie near a low-dimensional linear manifold; rapid prototyping. POD preprocessing can reduce input dimension by orders of magnitude and accelerate training dramatically.

When it hurts. Strongly nonlinear regimes (moving shocks, advecting fronts, bifurcations): POD truncation is inefficient, and the NN downstream has to re-learn a nonlinear correction.

A POD-augmented variant uses classical POD on the output with a [Lu et al. \(2021\)](#)-style branch that predicts modal amplitudes; a clean hybrid between linear reduced-basis methods and neural operators.

7.4 Normalisations and scaling

Nondimensionalisation. Normalise by free-stream velocity U_∞ , chord / diameter L , and reference pressure/temperature/density: $u \leftarrow u/U_\infty$, etc. This removes trivial variability and forces the network to learn the non-trivial part.

Reynolds / Favre decomposition. For stationary turbulent flows, split $\mathbf{U} = \bar{\mathbf{U}} + \mathbf{U}'$; feed the mean to one head, the fluctuations to another. For compressible, use Favre (density-weighted) decomposition ([Pope, 2000](#)), $\mathbf{U} = \tilde{\mathbf{U}} + \mathbf{U}''$ with $\tilde{\mathbf{U}} = \overline{\rho\mathbf{U}}/\bar{\rho}$.

Wall scaling. In wall-bounded flows, use wall units $y^+ = yu_\tau/\nu$, $u^+ = u/u_\tau$. A field expressed in wall units exposes the universal log-law and viscous-sublayer structure; a field in lab coordinates hides it. This is a classical trick; neural surrogates benefit from it too.

Log-transforms. For quantities spanning many orders of magnitude — dissipation ε , enstrophy, scalar gradient — fit in log-space: $\log \varepsilon$ rather than ε . Prevents the network from being dominated by outliers.

7.5 Helmholtz decomposition

Any velocity field on a simply connected domain admits

$$\mathbf{u} = \nabla\phi + \nabla \times \boldsymbol{\psi}, \quad \nabla \cdot (\nabla \times \boldsymbol{\psi}) = 0,$$

i.e. a potential plus a solenoidal part. In incompressible flow the potential part is null; feeding the solenoidal part directly, or parametrising the network output as $\nabla \times \boldsymbol{\psi}$, enforces $\nabla \cdot \mathbf{u} = 0$ *exactly*, not approximately.

8 Dual-head architectures: physical and spectral in parallel

A natural architectural idea, flagged as a question earlier, is to let the encoder see the field simultaneously in two representations — physical and spectral — and fuse them in the latent. We develop this now.

8.1 Architecture

Let $\mathcal{F}$ denote an (appropriate) spectral transform — FFT for periodic directions, wavelet for non-periodic, POD for data-driven. Define two encoders:

$$\begin{aligned} \mathcal{E}^{\text{spat}} : \mathbf{U}(\mathbf{x}, t) &\mapsto \mathbf{z}^{\text{spat}} \in \mathbb{R}^{L_{\text{spat}}}, \\ \mathcal{E}^{\text{spec}} : \mathcal{F}[\mathbf{U}](\mathbf{k}, \omega) &\mapsto \mathbf{z}^{\text{spec}} \in \mathbb{R}^{L_{\text{spec}}}, \end{aligned}$$

and a fusion map $z_{\text{dyn}} = \mathcal{F}_{\text{fuse}}(z^{\text{spat}}, z^{\text{spec}})$ — a small MLP, cross-attention, or gated combination. The decoder has two heads in symmetry:

$$\mathcal{D}^{\text{spat}} : z_{\text{dyn}} \mapsto \widehat{U}^{\text{spat}}(\mathbf{x}, t), \quad \mathcal{D}^{\text{spec}} : z_{\text{dyn}} \mapsto \widehat{\mathcal{F}[U]}(\mathbf{k}, \omega).$$

8.2 Losses

Three terms, each with a clear role.

(1) **Physical reconstruction.** $\mathcal{L}_{\text{rec}}^{\text{spat}} = \mathbb{E} \|U - \widehat{U}^{\text{spat}}\|^2$.

(2) **Spectral reconstruction.** $\mathcal{L}_{\text{rec}}^{\text{spec}} = \mathbb{E} \|\mathcal{F}[U] - \widehat{\mathcal{F}[U]}\|^2$, evaluated over a set of wavenumbers / frequencies / wavelet scales chosen by the user. Often done in log-space to equalise contributions across orders of magnitude in $E(k)$.

(3) **Primal–dual consistency.** $\mathcal{L}_{\text{cons}} = \mathbb{E} \|\mathcal{F}[\widehat{U}^{\text{spat}}] - \widehat{\mathcal{F}[U]}\|^2$.

The third term enforces that the two heads produce a *consistent* pair: the spectrum of the reconstructed spatial field matches the directly-predicted spectrum. Without this, the two heads drift apart. With it, we have the network equivalent of the consistency between primal and dual formulations of a PDE: it is the same object described two ways.

Fluid-mechanics reading

Why is this useful? Because neural networks have a documented *spectral bias* (Rahaman et al., 2019): they preferentially learn low-frequency content and systematically under-represent high k . The spectral head counteracts this: the loss $\mathcal{L}_{\text{rec}}^{\text{spec}}$ forces the network to reproduce $E(k)$ across the whole resolved range, including the high- k dissipation bump. Training under $\mathcal{L}_{\text{rec}}^{\text{spat}}$ alone gives a network that reconstructs vortex streets but blurs the dissipative scales. Adding the spectral head fixes this.

8.3 Variants worth exploring

- *Windowed / local spectra.* In wall-bounded flows, a global FFT is meaningless. Use local spectra in homogeneous sub-directions, or wavelet transforms.
- *Helmholtz-decomposition heads.* Two heads: one for the solenoidal component $\mathbf{u}_\psi$, one for potential $\nabla\phi$. Enforces incompressibility structurally.
- *Frequency-ladder heads.* One head per octave of scales. Enforces bandpass-clean reconstruction; echoes multi-resolution analysis in wavelets.
- *POD head.* Auxiliary head that predicts POD amplitudes of the field. Ties the network to a classical linear reduced basis, making comparisons cleaner.

Part III

Operator-network architectures

9 A guided tour with pros and cons

Seven families dominate current operator-learning research. All can be plugged into the framework of Part IV as the specific form of $\mathcal{E}$ or $\mathcal{D}$. They differ in what representation they assume, what geometry they handle, and how they scale.

9.1 DeepONet and its geometric extensions

Formulation. Two networks: a *branch* $b(\boldsymbol{\mu}, \mathbf{g})$ taking input functions/parameters to coefficient vector; a *trunk* $\tau(\mathbf{x}, t)$ taking output coordinates to basis-function vector. Prediction is a dot product:

$$\hat{U}(\mathbf{x}, t; \boldsymbol{\mu}, \mathbf{g}) = \sum_{k=1}^B b_k(\boldsymbol{\mu}, \mathbf{g}) \tau_k(\mathbf{x}, t).$$

Pros. Universal-approximation theorem (Chen and Chen, 1995; Lu et al., 2021). Geometry-agnostic via the trunk. Handles arbitrary query locations. The branch-trunk structure is the direct descendant of Galerkin projection with learned coefficients — familiar territory for this audience. Geom-DeepONet (He et al., 2024) augments the trunk with point-cloud geometry; Rabeh et al. (2025b) add a CNN branch for history.

Cons. The dot-product output is *low-rank*: only B degrees of freedom at any query point. High- k spectral content needs many basis functions. Training is fine; inference at a dense grid requires evaluating the trunk at every point.

Best for. Parametric PDEs with smooth solutions, geometry-varying problems, situations where you want differentiable continuous queries.

9.2 Fourier Neural Operator (FNO)

Formulation. Each layer: $v \mapsto \sigma(Wv + \mathcal{F}^{-1}(\mathcal{R}_\theta \mathcal{F}v))$ where $\mathcal{F}$ is FFT, $\mathcal{R}_\theta$ is a learned truncated low-pass tensor in k -space. Multiple layers compose.

Pros. Global receptive field in $\mathcal{O}(N \log N)$. Discretisation-invariant theoretical analysis. Captures long-range structure cheaply. Dominates on Darcy flow, homogeneous turbulence benchmarks. Mature library support.

Cons. Needs regular grid. Periodic-boundary assumption is baked in (padding workarounds exist). Complex geometry is painful: either parameter fields must be extended (masking), or one uses Geo-FNO (Li et al., 2023) which learns deformations. Limited to moderate dimensions (the spectral tensor grows).

Best for. Homogeneous or near-homogeneous problems; periodic domains; problems dominated by resolved large scales; fast training on modest meshes.

9.3 Convolutional Neural Operator (CNO)

Formulation. U-Net architecture (Ronneberger et al., 2015) with down-up convolutions, designed for discretisation-consistency: errors do not blow up as the grid is refined (Raonić et al., 2023).

Pros. Strong empirical performance on PDE benchmarks. Natural for 2D/3D fields on regular grids. Inherits the wealth of CNN tooling. Handles geometry variations through input masks (SDF or binary).

Cons. Grid-locked. Local convolution: long-range coupling comes from deep stacks of layers, which is a less efficient inductive bias than FNO’s spectral approach for some problems. Requires careful padding/grid alignment for variable geometries.

Best for. Regular-grid data with modest long-range coupling; industrial CFD outputs that already live on structured meshes; problems where geometry is captured via masks/SDFs rather than mesh deformation.

9.4 Graph Neural Operator (GNO) and MeshGraphNet

Formulation. The domain is a graph (mesh nodes with edges). Each layer is a message-passing step: every node aggregates from its neighbours through learned functions of relative position and node features.

Pros. Natively handles unstructured meshes, complex geometry, arbitrary topology. Local by design — maps onto finite-volume stencil updates. Strong for problems where locality dominates (shocks, contact lines, complex boundaries).

Cons. Long-range coupling needs many layers, which grows cost and training difficulty. Expensive at industrial mesh sizes (millions of cells). Permutation of node ordering is handled in principle but the design matters. Less mature than FNO/CNO at scale.

Best for. Unstructured CFD grids; aerospace / automotive geometries where remeshing is inevitable; problems where geometry varies freely.

9.5 Transformer-based operators

Formulation. Self-attention (Vaswani et al., 2017): for tokens $\{t_i\}$, $\text{Att}(Q, K, V) = \text{softmax}(QK^\top/\sqrt{d})V$. Tokens can be patches (vision transformer), mesh nodes, or modes. Scalable Operator Transformer (Poseidon; Herde et al., 2024) is a fluid-focused variant.

Pros. Captures long-range, data-dependent interactions — the natural inductive bias for triadic interactions in turbulence. Scales to large datasets. Supports foundation-model-style pre-training across many PDEs. Strongest empirical results on modern multi-PDE benchmarks (FlowBench; Tali et al., 2025).

Cons. Quadratic in sequence length for vanilla attention. Requires large training data. Geometry is handled via tokenisation, which is not automatic for arbitrary meshes.

Best for. Large, diverse training corpora; transfer-learning aspirations; problems with intrinsic long-range coupling (turbulence, acoustics).

9.6 Neural fields / Implicit neural representations (INR)

Formulation. Input: spatial (and temporal) coordinate. Output: field value. The function $U(\mathbf{x}, t; \theta)$ is represented *directly* by a coordinate-based MLP, possibly with sinusoidal SIREN activations (Sitzmann et al., 2020) or Fourier feature inputs (Tancik et al., 2020).

Pros. Continuous, resolution-free. Naturally handles arbitrary query locations. Compatible with SDF-based geometry (a single INR for geometry + one for the flow). Excellent for super-resolution (train on sparse, query on dense). This is the basis of Karki et al.’s INR + Shifted Boundary Method approach.

Cons. Each query is one forward pass: slow for dense grids. Training is slower than grid-based methods; convergence requires care (spectral bias (Rahaman et al., 2019) is severe for plain MLPs — hence SIREN (Sitzmann et al., 2020), Fourier features (Tancik et al., 2020)). Memory-resident representation per sample.

Best for. Geometry representation; super-resolution; uncertainty quantification via last-layer linearisation (Karki); problems where a mesh is a nuisance.

9.7 Diffusion-based operators

Formulation. Generate samples from $p(\mathbf{U} \mid \boldsymbol{\mu}, \mathbf{g})$ by iterative denoising from Gaussian noise, conditioned on operating/geometric parameters. PDE-Refiner (Lippe et al., 2023) refines an initial operator prediction; latent-diffusion models (Rombach et al., 2022; Nirmal et al., 2025) operate in compressed latent space.

Pros. Long-horizon stability (iterative refinement corrects accumulated errors). Natural stochastic modelling — ideal for chaotic flows where pointwise prediction is meaningless but ensembles are well-defined. Uncertainty quantification for free (sample from the posterior). Exceptional sample quality.

Cons. Slow inference (tens–hundreds of denoising steps). Requires careful noise schedule design. Theoretical analysis for operator learning is early-stage.

Best for. Chaotic / stochastic flows (turbulent inflow generation, LES realisations); ensemble forecasting; long rollouts where deterministic operators fail; probabilistic design exploration.

10 Recommendation matrix

A compact summary.

Architecture	Reg. grid	Geom.	Spec.	Data	Long range	Typical use
DeepONet	no	good	weak	moderate	moderate	Parametric smooth flows
FNO	yes	poor	strong	moderate	strong	Periodic turbulence
CNO (U-Net)	yes	via SDF	moderate	moderate	moderate	Industrial grid data
GNO / MeshGNN	no	strong	weak	high	weak	Unstructured CFD
Transformer	flex.	moderate	moderate	high	strong	Foundation-model scale
INR (SIREN,SDF)	no	strong	with FF	high	weak	Geometry, super-res
Diffusion	flex.	flex.	flex.	very high	strong	Chaotic, generative

MDO / computational-science reading

For most AutoDIDACTS-type problems, a sensible hybrid is: *GNO or INR encoder for geometry*, *FNO or spectral-head encoder for field content*, *DeepONet-style decoder for continuous query*, and *diffusion in latent space* for the stochastic / chaotic component. Each module is well-understood in isolation; the framework of Part IV below specifies how to glue them through a structured latent.

Part IV

The compositional framework

11 Latent-space architecture

We posit a *block-structured* latent variable

$$\mathbf{z} = [\mathbf{z}_\mu \parallel \mathbf{z}_g \parallel \mathbf{z}_\eta \parallel \mathbf{z}_\xi] \in \mathbb{R}^{L_\mu} \oplus \mathbb{R}^{L_g} \oplus \mathbb{R}^{L_\eta} \oplus \mathbb{R}^{L_\xi}, \quad (5)$$

with roles:

Block	Symbol	Content	Analogue
Parameter	$\mathbf{z}_\mu$	Re, M , forcing dependence	“environmental” (Powadi et al.)
Geometry	$\mathbf{z}_g$	shape / boundary descriptors	SDF trunk embed. (Rabeh et al.)
Coherent dynamics	$\mathbf{z}_\eta(t)$	time-evolving macroscale modes	TSD modes (Guo et al., 2016)
Residual	$\mathbf{z}_\xi(t)$	realisation-specific fluctuation	“specific” (Powadi et al., 2025)

Write $\mathbf{z}_{\text{stat}} = (\mathbf{z}_\mu, \mathbf{z}_g)$ for the time-invariant part and $\mathbf{z}_{\text{dyn}}(t) = (\mathbf{z}_\eta(t), \mathbf{z}_\xi(t))$ for the time-evolving part.

Fluid-mechanics reading

This partitioning is the latent-space analogue of a formal decomposition

$$\mathbf{U}(\mathbf{x}, t; \boldsymbol{\mu}, \mathbf{g}) = \underbrace{\overline{\mathbf{U}}(\mathbf{x}; \boldsymbol{\mu}, \mathbf{g})}_{\text{parametric mean}} + \underbrace{\sum_i a_i(t; \boldsymbol{\mu}, \mathbf{g}) \phi_i(\mathbf{x}; \mathbf{g})}_{\text{coherent modes}} + \underbrace{\mathbf{U}'(\mathbf{x}, t)}_{\text{residual}},$$

a generalised POD with mean subtraction in which mean, modes, and residual all depend on regime. The encoder learns this decomposition *nonlinearly*, and the disentanglement penalties (below) make the decomposition unique (up to discrete symmetries).

12 Three training paths

Path A — compositional autoencoder. Train $(\mathcal{E}, \mathcal{D})$ jointly with reconstruction, disentanglement, and (optional) physics losses. No explicit time propagator; queries at new $(\boldsymbol{\mu}, \mathbf{g})$ go $(\boldsymbol{\mu}, \mathbf{g}) \rightarrow \mathbf{z}_{\text{stat}} \rightarrow \hat{\mathbf{U}}$. Best when trajectories are short or fields are (quasi-)steady.

Path B — end-to-end propagator. Train $(\mathcal{E}, \mathcal{D}, \Phi)$ jointly, with $\Phi : \mathbf{z}_{\text{dyn}}(t) \mapsto \mathbf{z}_{\text{dyn}}(t + \Delta t)$ a latent time-stepper conditioned on $\mathbf{z}_{\text{stat}}$. Enables long rollouts but is harder to train.

Path C — two-stage (BLISS-style). Stage 1: train $(\mathcal{E}, \mathcal{D})$ as in Path A. Stage 2: freeze $(\mathcal{E}, \mathcal{D})$; train Φ on latent sequences $\{\mathbf{z}_{\text{dyn}}(t_k)\}$ alone. Matches the latent-diffusion recipe of [Nirmal et al. \(2025\)](#) and is typically the most stable.

MDO / computational-science reading

Path C is *identical in structure* to BLISS (Bi-Level Integrated System Synthesis; [Sobieszczanski-Sobieski et al., 2000](#)): solve local subproblems first, linearise / freeze them, then coordinate at system level. The empirical observation that Path C trains more stably than Path B is the operator-learning restatement of the classical MDO observation that bi-level architectures are more robust than monolithic ones. Path A + SINDy-on-latents = single-fidelity surrogate + symbolic correction — a workflow every MDO practitioner will recognise.

13 Compositional identifiability: group-structured minibatches

Disentanglement is *impossible* from unlabelled data alone ([Locatello et al., 2019](#)). Our inductive bias is the minibatch structure: we construct minibatches in which samples share factors. Specifically, every minibatch contains groups of samples with the same $\boldsymbol{\mu}$ but different $\mathbf{g}$, and groups with the same $\mathbf{g}$ but different $\boldsymbol{\mu}$. The Powadi-style fusion block then takes the encoded samples within a group and extracts a *shared* sub-code:

$$\mathbf{z}_{\mu}^{\star} = \text{Fuse}(\{\mathcal{E}(\mathbf{U}_i) : \boldsymbol{\mu}_i = \boldsymbol{\mu}^{\star}\}), \quad \mathbf{z}_{\xi}^{(i)} = \mathcal{E}(\mathbf{U}_i) - \Pi_{\mu} \mathbf{z}_{\mu}^{\star}.$$

This makes the factorisation identifiable up to trivial symmetries — a data-augmentation argument strongly analogous to that used in iVAE ([Khemakhem et al., 2020](#)).

14 Latent-space properties (desiderata)

The ten properties from our earlier note, here stated compactly and cross-referenced to the loss library of §15.

Property	Statement	Losses
P1 Compositionality	$\mathcal{Z} = \mathcal{Z}_{\mu} \oplus \mathcal{Z}_{\mathbf{g}} \oplus \mathcal{Z}_{\eta} \oplus \mathcal{Z}_{\xi}$; block-wise invariance.	L 8, L 10
P2 Disentanglement	$\mathbb{E}[\mathbf{z}_a \mathbf{z}_b^{\top}] - \mathbb{E}[\mathbf{z}_a] \mathbb{E}[\mathbf{z}_b]^{\top} = 0$ for $a \neq b$.	L 4, L 5, L 7
P3 Orthogonality	$\langle \mathbf{z}_a, \mathbf{z}_b \rangle = 0$ a.s.	L 6
P4 Equivariance	$\mathcal{E}(\mathcal{T}_h \mathbf{U}) = \tilde{\mathcal{T}}_h \mathcal{E}(\mathbf{U})$ for symmetries $\mathcal{T}_h$.	L 11, L 13
P5 Lipschitz	Encoder / decoder Lipschitz constants finite.	L 23
P6 Modal universality	Temporal covariance of $\mathbf{z}_{\eta}$ has stable eigenstructure.	L 18
P7 Energy cascade	PSD of decoded field matches ground truth.	L 19
P8 Sparsity	$\mathbb{E}[\ \mathbf{z}\ _0] \ll L$.	L 22
P9 Isometry	Euclidean $\ \mathbf{z}_i - \mathbf{z}_j\ _2$ tracks $\ \mathbf{U}_i - \mathbf{U}_j\ _{L^2}$.	L 20
P10 Calibrated UQ	$q(\mathbf{z} \mathbf{U})$ has posterior consistent with Gram spectrum.	L 21, L 24

15 Loss-function library

What follows is a catalogue of loss terms, organised by role. Each is stated as a mathematical object with its fluid-mechanical motivation. Weights $\{\lambda_i\}$ are not specified here — they are themselves objects of the multi-objective optimisation of §16.

15.1 Reconstruction fidelity

(L 1) L^2 field reconstruction. $\mathcal{L}_{\text{rec}} = \mathbb{E}[\|\mathbf{U} - \mathcal{D}(\mathcal{E}(\mathbf{U}))\|_{L^2}^2]$.

Fluid-mechanics reading

The energy spectrum $E(k) \sim k^{-5/3}$ in the inertial range (Kolmogorov K41) means that an L^2 budget is dominated by the energy-containing range; fine-scale turbulent fluctuations are under-weighted. $\mathcal{L}_{\text{rec}}$ is necessary for pointwise accuracy but insufficient for turbulence; pair with spectral (L 19) or structure-function (L 20) losses.

(L 2) Sobolev / derivative-consistency loss. $\mathcal{L}_{\text{sob}} = \mathbb{E}[\|\nabla \mathbf{U} - \nabla \hat{\mathbf{U}}\|^2 + \|\partial_t \mathbf{U} - \partial_t \hat{\mathbf{U}}\|^2]$.

Fluid-mechanics reading

Vorticity $\boldsymbol{\omega} = \nabla \times \mathbf{u}$, rate-of-strain $S_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$, enstrophy $\Omega = \frac{1}{2} \int |\boldsymbol{\omega}|^2$, dissipation $\varepsilon = 2\nu \int S_{ij} S_{ij}$, and wall shear stress $\tau_w = \mu(\partial u / \partial n)|_{\text{wall}}$ are all *gradient functionals* of $\mathbf{u}$. Two fields close in L^2 can be wildly different in their derivatives (a wiggly sawtooth vs. its mean). [Rabeh et al. \(2025b\)](#) show derivative losses are decisive for drag and Strouhal-number fidelity.

(L 3) Boundary-layer weighted loss. $\mathcal{L}_{\text{bdry}} = \mathbb{E}[\int w(\phi_g(\mathbf{x})) \|\mathbf{U} - \hat{\mathbf{U}}\|^2 d\mathbf{x}]$, $w(s) = e^{-s^2/\delta^2}$.

Fluid-mechanics reading

The viscous sublayer $y^+ \lesssim 5$ is $\sim 10^{-4}$ of the boundary-layer thickness. A uniform L^2 norm treats near-wall and free-stream errors identically, yet near-wall errors determine skin friction, heat transfer, and transition. Weighting by a tubular SDF-based kernel concentrates the training signal where it matters.

(L 4) Masked reconstruction for gappy data. $\mathcal{L}_{\text{mask}} = \mathbb{E}[\frac{1}{|\mathcal{O}_i|} \sum_{m \in \mathcal{O}_i} \|\mathbf{U}_m - \hat{\mathbf{U}}(\mathbf{x}_m, t_m)\|^2]$.

Fluid-mechanics reading

Directly extends $\mathcal{L}_{\text{rec}}$ to handle experimental data (PIV, LDV, pressure taps) where only some locations are observed. Operator-learning analogue of gappy POD (Everson & Sirovich 1995).

(L 5) Heteroscedastic (noise-weighted) loss. $\mathcal{L}_{\text{het}} = \mathbb{E}[\sum_m \frac{\|\mathbf{U}_m - \hat{\mathbf{U}}(\mathbf{x}_m, t_m)\|^2}{2\sigma_m^2} + \log \sigma_m]$.

Fluid-mechanics reading

Equivalent to weighted least-squares with observation covariance $R_{mm} = \sigma_m^2$. Identical struc-

ture to the observation term in 3DVAR / 4DVAR data assimilation. The $\log \sigma_m$ term prevents the trivial solution $\sigma_m \rightarrow \infty$.

15.2 Disentanglement and orthogonality

(L 6) Pearson correlation penalty (Powadi et al., 2025). $\mathcal{L}_{\text{corr}} = \sum_{a \neq b, i \in a, j \in b} |\text{Corr}(z_i, z_j)|$.

Fluid-mechanics reading

In turbulent flows, Re governs the balance of inertia and viscosity; M controls compressibility. Their effects are *approximately* orthogonal away from the transonic regime. Decorrelated latent coordinates enable clean extrapolation along each axis.

(L 7) HSIC (nonlinear independence). $\mathcal{L}_{\text{HSIC}} = \sum_{a < b} \text{HSIC}(z_a, z_b; k_a, k_b)$.

Fluid-mechanics reading

Many fluid interactions are nonlinear: transonic drag divergence, roughness-Reynolds coupling, compressibility effects on skin friction. HSIC detects any functional dependence, not just linear correlation.

(L 8) Block-Gram orthogonality. $\mathcal{L}_{\text{ortho}} = \sum_{a \neq b} \|\mathbb{E}[z_a z_b^\top]\|_F^2$.

Fluid-mechanics reading

The latent-space analogue of POD’s spatial-mode orthogonality: variance decomposes cleanly across blocks with no cross-terms, exactly as POD energy decomposes across modes. Makes downstream SINDy regression well-conditioned.

(L 9) Mutual-information minimisation. $\mathcal{L}_{\text{MI}} = \sum_{a \neq b} \text{KL}(q(z_a, z_b) \| q(z_a)q(z_b))$.

Fluid-mechanics reading

Strongest form of disentanglement. Essential when parameter regimes span qualitatively different physics (attached vs. separated, laminar vs. turbulent, subsonic vs. supersonic).

15.3 Compositional identifiability

(L 10) Same-factor invariance. For (i, j) with $\mu_i = \mu_j$: $\mathcal{L}_{\text{inv}}^\mu = \mathbb{E}[\|z_\mu^{(i)} - z_\mu^{(j)}\|^2]$.

Fluid-mechanics reading

Articulates *universality*: different realisations at the same regime over the same shape must map to the same point in z_μ . Ensemble DNS/LES runs are the natural training data.

(L 11) Contrastive (InfoNCE). $\mathcal{L}_{\text{NCE}}^a = -\log \frac{\exp(\text{sim}(z_a^{(i)}, z_a^{(j)})/\tau)}{\sum_k \exp(\text{sim}(z_a^{(i)}, z_a^{(k)})/\tau)}$.

Fluid-mechanics reading

Endows latent with a similarity metric in which same-regime samples are close and different-regime are far. Analogue of turbulence regime clustering.

(L 12) Swap consistency. $\mathcal{L}_{\text{swap}} = \mathbb{E}[\|U^{(\mu_i, g_j)} - \mathcal{D}(z_\mu^{(i)}, z_g^{(j)}, \cdot, \cdot)\|^2].$

Fluid-mechanics reading

Strongest compositional loss: take μ -embedding of sample i , g -embedding of sample j , concatenate, decode, compare to ground truth at the cross-combination. If this works, the factorisation is *functional*, not merely statistical — a strong demand akin to Buckingham-Pi consistency across geometries at fixed dynamic similarity.

(L 13) Cross-source consistency. $\mathcal{L}_{\text{src}} = \|\mathcal{E}^{\text{DNS}}(U_i^{\text{DNS}}) - \mathcal{E}^{\text{exp}}(U_i^{\text{exp}})\|^2.$

Fluid-mechanics reading

When the same (μ_i, g_i) exists in both DNS and experiment, embeddings must agree. Ties multi-source training together and prevents encoder drift.

15.4 Dynamics and rollout

(L 14) Single-step latent loss. $\mathcal{L}_{\text{step}} = \mathbb{E}[\|\Phi_{\Delta t}(z_{\text{dyn}}(t)) - \mathcal{E}^U(U(\cdot, t + \Delta t))\|^2].$

Fluid-mechanics reading

One latent CFL step. For vortex-shedding, Δt should resolve the Strouhal period; a rule of thumb is $\Delta t \sim T_s/20$.

(L 15) k -step pushforward. $\mathcal{L}_{\text{roll}}^{(K)} = \sum_{k=1}^K \gamma^k \mathbb{E}[\|\mathcal{D}(\Phi_{\Delta t}^k(z_{\text{dyn}}(0)); \mathbf{x}, t_k) - U(\mathbf{x}, t_k)\|^2].$

Fluid-mechanics reading

Combats exposure bias: model sees its own rollouts. For chaotic flows, pointwise long-horizon prediction is impossible (Lyapunov divergence); what is achievable is *statistical* long-horizon accuracy — correct shedding frequency, integral time scale.

(L 16) Static invariance. $\mathcal{L}_{\text{stat}} = \mathbb{E}[\|z_{\text{stat}} - z_{\text{stat}}(t_k)\|^2].$

Fluid-mechanics reading

Re and g don't evolve; the propagator must not covertly drift them.

(L 17) Koopman spectral constraint. $\mathcal{L}_{\text{koop}} = \|\Phi V - V \text{diag}(\lambda)\|_F^2.$

Fluid-mechanics reading

Pushes Φ toward linearity on a Koopman-invariant subspace. If achieved, rollout is matrix exponentiation, and spectral analysis of Φ yields physical modes — the direct connection to DMD (Schmid, 2010; Rowley et al., 2009).

15.5 Physics-informed terms

(L 18) **PDE residual** (cf. PINNs; Raissi et al., 2019). $\mathcal{L}_{\text{phys}} = \mathbb{E}[\|\mathcal{N}[\widehat{U}](\mathbf{x}, t; \boldsymbol{\mu})\|^2 + \|\nabla \cdot \widehat{U}\|^2]$.

Fluid-mechanics reading

Direct injection of governing equations. For incompressible NS: $\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p - \nu \nabla^2 \mathbf{u} = 0$, $\nabla \cdot \mathbf{u} = 0$. **Warning:** if training data are LES/RANS-filtered, using the unfiltered residual implicitly assumes a zero subgrid term — a silent closure assumption.

(L 19) **Conservation of first integrals.** $\mathcal{L}_{\text{cons}} = \sum_I (I[\widehat{U}] - I[U])^2$, $I \in \{\text{mass, momentum, energy, enstrophy, h}\}$.

(L 20) **Boundary condition loss.** $\mathcal{L}_{\text{bc}} = \mathbb{E}_{\mathbf{x} \in \partial\Omega_g} \|\widehat{U}(\mathbf{x}, t) - U_{\text{bc}}(\mathbf{x}, t)\|^2$.

15.6 Statistical / spectral losses

(L 21) **Temporal covariance matching.** $\mathcal{L}_{\text{cov}} = \|C_{\widehat{U}}(t, t') - C_U(t, t')\|_F^2$.

Fluid-mechanics reading

Encodes all linear time scales: integral time $T_L = \int R(\tau) d\tau$, decorrelation times, memory. The building block of the TSD framework of Guo et al. (2016).

(L 22) **Power-spectrum matching (log-space).** $\mathcal{L}_{\text{psd}} = \int |\log E_{\widehat{U}}(k) - \log E_U(k)|^2 dk + (\text{temporal})$.

Fluid-mechanics reading

Directly addresses spectral bias of plain MLPs. Enforces K41 scaling in the inertial range (see e.g. Pope, 2000), captures dissipative bump near k_η . Indispensable when the spectral head of §8 is in play.

(L 23) **Higher-order structure functions.** $\mathcal{L}_{\text{struct}}^{(p)} = |S_p^{\widehat{U}}(r) - S_p^U(r)|^2$, $S_p(r) = \mathbb{E}[|\delta U(r)|^p]$.

Fluid-mechanics reading

Captures *intermittency*. The anomalous scaling ζ_p of $S_p(r) \sim r^{\zeta_p}$ deviates from K41's $p/3$ for $p \gtrsim 4$ in real turbulence — a signature of rare extreme events (Pope, 2000). MSE-only training gives near-Gaussian statistics and flat ζ_p ; this loss restores non-Gaussian intermittency.

15.7 Regularisation and compactness

(L 24) **KL to prior (VAE; Kingma and Welling, 2014).** $\mathcal{L}_{\text{KL}} = \text{KL}(q(\mathbf{z}|U) \| p(\mathbf{z}))$.

(L 25) **Sparsity.** $\mathcal{L}_{\text{sp}} = \mathbb{E}[\|z\|_1]$.

(L 26) **Lipschitz (spectral-norm) penalty.** $\mathcal{L}_{\text{lip}} = \sum_{\ell} \sigma_{\max}(W^{(\ell)})$.

Fluid-mechanics reading

Bounded Lipschitz is the operator-learning analogue of CFL stability: global rollout error scales as L_{Φ}^K , so $L_{\Phi} > 1$ destabilises long horizons.

(L 27) **Eikonal regulariser.** $\mathcal{L}_{\text{eik}} = \mathbb{E}[(\|\nabla\phi\| - 1)^2]$.

Fluid-mechanics reading

Enforces ϕ is a true SDF. Karki et al. (2025a) show this stabilises the downstream BVP solve: small level-set errors produce proportionally small boundary errors.

Part V

Optimisation perspectives

16 Multi-objective optimisation

The training objective has many parts that are not directly commensurate in units, magnitude, or noise. The natural formulation is

$$\min_{\Theta} (\mathcal{L}_1(\Theta), \mathcal{L}_2(\Theta), \dots, \mathcal{L}_M(\Theta)),$$

for which no single “best” exists; instead there is a Pareto set $\{\Theta^* : \nexists \Theta' \text{ with } \mathcal{L}_i(\Theta') \leq \mathcal{L}_i(\Theta) \forall i, \text{ strict for some } i\}$.

16.1 Classical scalarisations

- *Weighted sum:* $\sum_i \lambda_i \mathcal{L}_i$ (see e.g. Miettinen, 1999). Simplest; recovers convex Pareto portions only.
- *Chebyshev:* $\max_i \lambda_i (\mathcal{L}_i - \mathcal{L}_i^*)$. Recovers non-convex fronts too.
- ε -*constraint:* $\min \mathcal{L}_1$ s.t. $\mathcal{L}_i \leq \varepsilon_i$. Good when some losses are hard tolerances.
- *Achievement scalarising function:* $\max_i \lambda_i (\mathcal{L}_i - r_i) + \rho \sum_i \lambda_i \mathcal{L}_i$.

16.2 Adaptive weighting

- *GradNorm* (Chen et al., 2018b): rescale λ_i so per-task gradient norms balance.
- *Uncertainty weighting* (Kendall et al., 2018): $\lambda_i = 1/(2\sigma_i^2)$ with σ_i learned; gives $\mathcal{L} = \sum_i \mathcal{L}_i / (2\sigma_i^2) + \log \sigma_i$.
- *PCGrad* (Yu et al., 2020) and *MGDA* (Désidéri, 2012): project or combine gradients to avoid destructive interference between tasks.

16.3 Pareto-ensemble deployment

Rather than committing to one λ , train a family $\{\Theta^*(\lambda^{(i)})\}$ sweeping the Pareto front. At deployment, the user (or outer optimiser) selects the model best suited to each query — the low- $\mathcal{L}_{\text{psd}}$ variant for spectral queries, the low- $\mathcal{L}_{\text{bdry}}$ variant for skin-friction, and so on. Hypernetwork-based conditional training amortises the cost.

MDO / computational-science reading

This is exactly robust design under preference uncertainty: rather than choose one design, present the Pareto front and let the decision-maker select. In MDO, this is a standard step; in operator learning, it is still novel.

17 Multidisciplinary design optimisation

Our treatment follows the MDO-architecture taxonomy of [Martins and Lambe \(2013\)](#).

Two linkages, kept separate.

17.1 Training as MDO

Treat each latent block as a “discipline” with its own losses. The shared latent dimensions are coupling variables.

MDO architecture	Training-loop analogue
MDF (Multi-Discipline Feasible)	Joint end-to-end (Path B): every gradient step satisfies all blocks’ losses.
IDF (Individual Discipline Feasible)	Alternating optimisation; coupling via shared latent dims.
CO (Collaborative Optimisation)	Bi-level: each block has local subproblem; system level enforces consistency.
BLISS	Train locally, linearise / freeze, then system-level coordination. = Path C.
AAO (All-At-Once)	Discretise latent PDE jointly with autoencoder.

17.2 Surrogate as MDO building block

Once trained, the operator replaces CFD inside any MDO loop:

- *Aero-structural*. Surrogate supplies pressure and wall-shear on $\partial\Omega_g$; structural solver consumes them. Autodiff gives adjoint sensitivities through both surrogates for free.
- *Aero-thermal*. Extend with temperature channel; couple to thermal-conduction solver. Hypersonic reentry, turbine cooling.
- *Aero-control co-design*. Controller acts via μ ; closed-loop latent propagator is a reduced-order controlled-flow model.
- *Multi-fidelity hierarchy*. CFD ground truth (slow, authoritative), learned operator (fast, differentiable, calibrated), coarse closed-form surrogate (fastest, roughest): fused via multi-fidelity Monte Carlo, hierarchical GPs, co-Kriging.

17.3 Disentanglement as design-space conditioning

Classical MDO suffers from ill-conditioned design landscapes: correlated variables yield curved Pareto fronts and degenerate gradients. Disentangled latents provide an *intrinsically orthogonal* design parameterisation: moving along $\mathbf{z}_\mu$ changes regime, moving along $\mathbf{z}_g$ changes shape, and these do not interfere. This is the latent-space analogue of automatically finding a good shape-design basis.

17.4 Robust / reliability-based MDO

With probabilistic encoders (VAE, Karki-Gram last-layer), each prediction carries a calibrated variance. The MDO formulations

$$\begin{aligned} \text{Robust: } & \min_{\boldsymbol{\mu}, \mathbf{g}} \mathbb{E}[\Psi(\hat{\mathbf{U}})] + \kappa \text{Var}[\Psi(\hat{\mathbf{U}})], \\ \text{Reliability: } & \min_{\boldsymbol{\mu}, \mathbf{g}} \Psi(\hat{\mathbf{U}}) \text{ s.t. } \Pr[g_j(\hat{\mathbf{U}}) > 0] \leq p_j, \end{aligned}$$

with expectations over the latent posterior, become tractable. The uncertainty lives in a low-rank Gram-eigenmode subspace (Karki et al., 2025a), making sampling cheap.

17.5 Multi-fidelity fusion

The disentangled latent is a natural fusion target:

$$\mathbf{z}_{\text{HF}} = \mathbf{z}_{\text{LF}} + \Delta_\theta(\mathbf{z}_{\text{LF}}, \boldsymbol{\mu}, \mathbf{g}),$$

with $\mathbf{z}_{\text{LF}}$ from a large low-fidelity corpus and Δ_θ a small correction trained on sparse high-fidelity data. Direct analogue of co-Kriging’s additive bias correction (Kennedy and O’Hagan, 2000; Peherstorfer et al., 2018).

Part VI

Use and open questions

18 Downstream use

1. **Rapid design-space exploration.** Encode $(\boldsymbol{\mu}^*, \mathbf{g}^*)$, roll out, decode. Speed-ups of several orders of magnitude over direct CFD.
2. **Symbolic discovery.** Apply SINDy (Brunton et al., 2016; Champion et al., 2019) to latent trajectories, stratified by $(\boldsymbol{\mu}, \mathbf{g})$:

$$\frac{d\mathbf{z}_\eta}{dt} = \sum_j c_j(\boldsymbol{\mu}, \mathbf{g}) \Theta_j(\mathbf{z}_\eta).$$

Orthogonality (P3) conditions the regression; sparsity (P8) makes the output parsimonious; Koopman-linearity (L 17) collapses SINDy to eigendecomposition.

3. **Multi-fidelity fusion.** As §17.5.
4. **Uncertainty quantification.** Probabilistic encoder or Karki-Gram last-layer ensemble, propagated through Φ and $\mathcal{D}$ to produce calibrated distributional predictions.
5. **Active / dynamic sampling.** Use predicted variance as acquisition; expected variance reduction, expected improvement, or BALD all work on latent quantities.
6. **Inverse problems.** For $\mathbf{U}^{\text{obs}}$, solve $(\boldsymbol{\mu}^*, \mathbf{g}^*) = \arg \min \|\mathcal{E}^U(\mathbf{U}^{\text{obs}}) - \text{propagate}(\mathcal{E}^{\mu g}(\boldsymbol{\mu}, \mathbf{g}))\|^2$ — differentiable via autodiff.
7. **Transfer across regimes.** Disentangled $\mathbf{z}_\mu$ makes extrapolation in operating conditions well-defined.
8. **Generative trajectory sampling.** Latent diffusion (Nirmal et al., 2025) for plausible realisations: ensemble load estimates, turbulent-inflow generation, ensemble forecasting.
9. **Anomaly / regime-change detection.** Spikes in $\|\mathbf{z}_\xi\|$ or spectral shifts in $\mathbf{z}_\eta$ flag off-manifold inputs: transition, separation, shock formation.
10. **Data-driven closure.** The residual $\mathbf{z}_\xi$ captures what the mean-field model misses; a stochastic map $\mathbf{z}_\xi(t + \Delta t) \mid \mathbf{z}_\eta(t), \boldsymbol{\mu}$ is a learned subgrid closure.

19 Practical implementation

All quantitative choices below are determined at training time rather than pre-specified:

- *Latent-block dimensions.* Cross-validation on held-out $(\boldsymbol{\mu}, \mathbf{g})$ with an Occam prior favouring smaller blocks until reconstruction quality degrades. Bayesian model comparison over block sizes is an alternative.
- *Loss weights λ_i .* Adaptive (GradNorm, uncertainty weighting, MGDA), or swept in a Pareto-ensemble.
- *Training schedule.* Warm-up on reconstruction; phase in disentanglement and compositional losses; phase in dynamics losses with progressive rollout horizon; phase in physics and spectral losses for final polish.
- *Batch composition.* Stratified groups sharing $\boldsymbol{\mu}$ or $\mathbf{g}$, to make $\mathcal{L}_{\text{inv}}$, $\mathcal{L}_{\text{swap}}$, $\mathcal{L}_{\text{src}}$ well-defined.
- *Diagnostics.* Silhouette of $\mathbf{z}_\mu$ clustered by regime; covariance-spectrum match (P6); PSD comparison (P7); rollout divergence-norm and phase error (Rabeh et al., 2025b).

20 Open questions

- *Identifiability theory.* Can the group-structured minibatching of §13 be formalised into an identifiability theorem, generalising iVAE (Khemakhem et al., 2020) to the operator-learning setting?
- *Koopman linearity.* When (if ever) is Φ approximately linear on a Koopman-invariant subspace of $\mathcal{Z}$? Empirical tests on canonical flows before committing to theory.

- *Orthogonality vs. disentanglement.* For chaotic flows, residual $\mathbf{z}_\xi$ is genuinely correlated with shedding events in $\mathbf{z}_\eta$. Hard orthogonality may hurt expressivity; soft block-Gram may be right.
- *Gram-spectrum $\leftrightarrow$ block correspondence.* Is there a natural correspondence between Karki et al. (2025a)’s Gram eigenmodes and the compositional blocks? A positive result would make Path A’s disentanglement provable from spectral structure rather than merely enforced through losses.
- *Conditional-entropy sizing of L_ξ .* Tie the residual-block dimension to $H(\mathbf{U} \mid \boldsymbol{\mu}, \mathbf{g})$, the residual entropy. Measurable experimentally for turbulent systems.
- *MDO–MOO duality in training.* Make precise the observation that Path C’s stability is the operator-learning restatement of BLISS’s robustness, by mapping Path C’s hyperparameters onto BLISS coordination parameters.
- *Spectral–physical head ratio.* In the dual-head architecture of §8, what is the optimal $L_{\text{spat}}/L_{\text{spec}}$ ratio for a given spectral bandwidth? An empirical question with theoretical consequences.

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